

Vertex Example

The diagram shows the sum of three Feynman diagrams representing a vertex correction, which equals zero. The first diagram shows a fermion line with a vertex correction loop (a triangle with a dashed line) and a W boson line. The second diagram shows the same loop but with the W boson line attached to the other vertex. The third diagram shows a four-point vertex with two fermion lines and two W boson lines. The sum of these diagrams is zero, representing the Ward identity.

$$\text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} = 0$$